TRIzol-based RNA Extraction for miRNA Retention

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SUSTech-BIO 2026 iGEM Team

Purpose

Extract total RNA, including small RNA, from adherent cultured cells for subsequent miRNA analysis. This guide adapts the earlier M0 Day 2 extraction workflow and ends at RNA quality control and storage. It does not establish quantitative recovery of every miRNA. [1]

The increased isopropanol volume, overnight precipitation at -20°C, 30 min pellet spin and two ethanol washes are retained from that workflow. Overnight freezing is an adaptation, not the manufacturer's routine precipitation condition; assess salt carryover before downstream use. [1, 3]

Materials & Equipment

Item	Manufacturer	Identifier / specification	Use / Specification
TRIzol Reagent	Invitrogen	15596018	1 mL per well of a 6-well plate
Chloroform; isopropanol	Sangon Biotech	Molecular biology / analytical grade	Analytical / molecular biology grade; 0.2 mL chloroform per 1 mL TRIzol
Absolute ethanol	Sangon Biotech	Absolute grade	Prepare 75% ethanol (v/v) with DEPC-treated water
DEPC-treated water	Sangon Biotech	B501005	RNase-free water for ethanol preparation, RNA dissolution, dilution and measurement blank [5]
PBS	Gibco	Sterile phosphate-buffered saline	One brief wash before lysis in the adapted adherent-cell workflow
Pipette tips	Axygen tips; laboratory equipment	RNase-free, volume-compatible tips	RNase-free tips for RNA handling
RNase-free tubes; pipettes	Laboratory supplies	RNase-free tubes; calibrated pipettes	Solvent-compatible polypropylene tubes rated for 12,000 × g; allow mixing headspace

Item	Manufacturer	Identifier / specification	Use / Specification
Refrigerated microcentrifuge	Eppendorf centrifuge; laboratory equipment	Refrigerated; compatible rotor	Compatible rotor; 12,000 × g and 7,500 × g at 4°C
Microvolume spectrophotometer	Thermo Scientific	NanoDrop One	RNA concentration, absorbance spectrum and purity ratios
Heat block / water bath	As available	Temperature-controlled heating	55°C for RNA dissolution
Chemical fume hood; freezers	As available	Chemical ventilation; cold storage	Solvent handling; -20°C precipitation and -80°C RNA storage

Before You Start

Use the original working scale: one well of adherent cells in a 6-well plate, approximately 1 × 10^6 cells, with 1 mL TRIzol. Keep cell input and processing consistent across samples. This carrier-free workflow has not been established for low-input specimens. [1]

Precool the centrifuge to 4°C and prepare 75% ethanol in DEPC-treated water. Plan overnight precipitation followed by next-day pelleting. Set the dissolution block to 55°C. [1]

Handle TRIzol and chloroform in a chemical fume hood with appropriate gloves, eye protection and a lab coat; collect solvent-containing waste in the designated chemical-waste stream. Use the actual product SDS and local disposal requirements.

Select tubes by total liquid volume and rotor rating. With 0.60 mL aqueous phase plus 0.90 mL isopropanol, a 1.5 mL tube leaves no mixing headspace; use a compatible 2 mL tube for that volume.

Procedure

Cell lysis

1. Remove culture medium. Briefly rinse the adherent cells once with cold PBS, remove the wash completely, and proceed immediately to lysis without leaving the cells dry. This wash is retained from the earlier adherent-cell protocol. [1]

2. Add 1 mL TRIzol directly to the well. Pipette across the surface until the cells are dispersed and no visible clumps remain; transfer the lysate to a labeled tube. Leave at room temperature for 5 min. [1]

Phase separation

3. Add 0.2 mL chloroform per 1 mL starting TRIzol. Close securely and shake vigorously by hand for 15 s; leave at room temperature for 3 min. [1]

4. Centrifuge at 12,000 × g for 15 min at 4°C. Identify the upper aqueous phase, the intervening interface and the lower organic phase. [1]

5. Transfer the clear upper phase into a fresh tube, stopping before the interface. Record the volume actually collected, commonly around 0.5 mL at this scale; do not force recovery of a preset volume. [1]

RNA precipitation

6. Add 1.5 volumes of isopropanol relative to the recovered aqueous phase and mix by inversion. For 0.50 mL aqueous phase, add 0.75 mL isopropanol. The denominator is the aqueous-phase volume, not the original TRIzol volume. [1]

This isopropanol increase is compatible with the manufacturer's small-RNA guidance, which allows testing up to 1.0 mL per 1 mL TRIzol while checking salt carryover. Do not extrapolate the ratio beyond that range without reassessment. [3]

7. Incubate at -20°C overnight, retaining the earlier adapted condition. The routine manufacturer protocol instead uses 10 min at room temperature; cold overnight precipitation can increase salt co-precipitation and is not evidence of improved miRNA recovery by itself. [1–3]

8. Centrifuge at 12,000 × g for 30 min at 4°C. Locate the pellet, which may be small or difficult to see. Aspirate the supernatant from the opposite side without disturbing it. [1]

Ethanol washes and dissolution

9. Add 1 mL chilled 75% ethanol per 1 mL starting TRIzol and gently resuspend the pellet. Centrifuge at 7,500 × g for 5 min at 4°C, then remove the wash. Repeat once for two washes in total. [1]

10. Remove residual ethanol without touching the pellet. Air-dry for approximately 5–10 min; avoid prolonged drying until the pellet becomes hard and difficult to dissolve. [1]

11. Add 30 µL DEPC-treated water. Resuspend gently and incubate at 55°C for 10 min; mix again and check that the RNA has dissolved. Keep on ice for immediate measurements. [1]

Quality control and storage

12. Blank the NanoDrop One with the same DEPC-treated water used for dissolution. Measure the dissolved sample and save concentration, A260/A280, A260/A230 and the absorbance spectrum. Record any dilution factor and dissolution volume. [1, 4]

An A260/A280 near 2 and A260/A230 around 2.0–2.2 are useful reference values, not universal pass/fail limits. A low ratio can reflect contaminants or measurement conditions; it does not identify a single contaminant. Inspect the spectrum and repeat after checking the blank and pedestal. [4]

13. Prepare a working aliquot at a concentration suitable for the selected miRNA RT assay, using volumes within the pipette's reliable range. Keep the stock RNA at -80°C and minimize repeated thawing. Use a validated miRNA-compatible RT procedure and include no-RT and no-template controls in downstream testing. [1]

p.s.

Protect the interface margin. A slightly smaller clean aqueous transfer is preferable to carrying visible interface into the RNA tube. Keep the same handling order across samples.

The second ethanol wash addresses potential carryover after increased isopropanol and cold precipitation; it does not guarantee removal of all salts. If a pause is needed after pelleting, the manufacturer allows storage under 75% ethanol at -20°C for up to one year. [3, 2]

Do not substitute a generic cleanup column without confirming mature-miRNA retention. "Total RNA" alone is not sufficient evidence that a cleanup method preserves the relevant size range.

A late miRNA or U6 Cq is not a diagnosis of extraction failure. Expression, RNA input, RT performance, inhibition and assay specificity also matter. U6 detection does not demonstrate unbiased recovery of mature miRNAs.

A dilution series prepared from finished cDNA tests the PCR stage; it cannot rule out inhibition that occurred during reverse transcription. Investigate suspected RT inhibition with RNA inputs varied before RT while keeping the assay design controlled.

Expected Outcome

Obtain dissolved total RNA with documented concentration, spectrum and purity ratios. Pellet visibility and total RNA yield vary with input and do not quantify the small-RNA fraction.

Suitability for miRNA work requires downstream assay evidence with appropriate controls. Do not impose a universal U6 Cq, RNA concentration or recovery yield as proof of success.

Troubleshooting

Observed problem	Check first	Action
Poor separation or visible material in the upper layer	Mixing, reagent volumes, input load and centrifuge temperature	Confirm the chloroform ratio and separation settings. Transfer only a clean aqueous layer; avoid carrying the interface forward.
Small / invisible pellet or low yield	Input, aqueous volume, precipitation ratio and pellet handling	Review the recorded volumes and actual × g. Handle the expected pellet location carefully. Low-input samples need a separately established carrier-assisted method.
Low purity ratios or unexpected spectral shape	Blank, pedestal, full dissolution and solvent / salt carryover	Repeat the measurement correctly. Review both washes and phase transfer. Persistent contamination requires cleanup demonstrated to retain small RNA, or a fresh extraction. [4]
Pellet does not dissolve	Overdrying and dissolution conditions	Repeat gentle mixing within the 55–60°C, 10–15 min dissolution conditions; verify dissolution before quantification. [2]
Weak or inconsistent downstream signal	Input, assay controls, RNA-to-RT dilution and specificity	Separate extraction loss, RT inhibition and PCR failure using controlled comparisons. Do not diagnose missing miRNA from one Cq cutoff.
Signal in a no-RT control	Interface carryover, assay design and contamination	Assess residual DNA and setup contamination. If needed, use RNase-free DNA removal with cleanup verified to retain small RNA before repeating RT.

Reference

[1] Project source: M0D2_miRNA_stemloop_protocol-5.pdf, pp. 1–5, especially sections 3 and 5.1–5.3. Maintained source: GFA+MRE/M0/. Reopened 2026-10-03. Source of the adapted extraction conditions.

[2] Thermo Fisher Scientific. TRIzol Reagent User Guide, MAN0001271, Rev. D, 31 March 2025. RNA workflow and quantification, pp. 1–3. Accessed 2026-10-03. Open source

[3] Thermo Fisher Scientific. TRIzol Reagent FAQs: small-RNA precipitation; stopping points and storage. Increased isopropanol, salt carryover and a second ethanol wash. Accessed 2026-10-03. Open source

[4] Thermo Scientific. Assessment of Nucleic Acid Purity, technical note TN52646, pp. 1–3. Purity ratios, contaminants and measurement interpretation. Accessed 2026-10-03. Open source

[5] Sangon Biotech. B501005 DEPC-treated water, SDS, revision 2024-12-24; product identity. Accessed 2026-10-03. Open source